FEDERAL SPECIFICATION

PIPET, VOLUMETRIC; MICRO, LANG-LEVY DESIGN

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers Lang-Levy design volumetric pipets of micro volumes designated as microliters (μl).

1.2 Classification. Micropipets shall be of the following types and capacities, as specified (see 6.2).

1.2.1 *Types.*

Type I—Borosilicate glass.

Type II—Soda-lime glass.

1.2.2 *Capacities (microliters).*

1	15	50	200
2	20	60	250
3	25	75	300
5	30	100	400
10	40	150	500

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

DD-G-541—Glass (Laboratory).

UU-T-111—Tape; Paper, Gummed (Sealing and Securing).

NNN-B-789—Buret, Straight, Precision.

PPP-B-566—Boxes, Folding, Paperboard.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

PPP-B-676—Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

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Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Freight Classification Rules.

(Application for copies of these freight classification rules should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

American Society for Testing and Materials:

C148—Polarscopic Examination of Glass Containers

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Type I.* Type I Lang-Levy design micropipets shall be made from borosilicate glass conforming to type I, class A or B of DD-G-541.

3.1.2 *Type II.* Type II Lang-Levy design micropipets shall be made from soda-lime glass conforming to type II of DD-G-541.

3.2 Annealing. Ring strain shall be no greater than Temper No. 1 when tested as specified in 4.6.1. Any longitudinal strain, if present, shall be faint and highly diffused.

3.3 Capacity. Capacity (see 3.4.1) at 20°C, shall be a single volume within a tolerance of plus or minus 5 percent when delivered as specified in 4.6.2.

3.4 Shape. The shape shall be as illustrated in figure 1 (see 6.4); dimensions, where specified, shall be as shown. The stem of each pipet shall be gradually and uniformly tapered for a distance of 40 to 55 millimeters (mm) above the tip bend. The top shall be gradually and uniformly tapered for a distance of 15 to 25 mm to approximately 3 mm outside diameter (OD) at the end. Termination of the top shall be perpendicular to the longitudinal axis of the micropipet and fire polished.

3.4.1 *Capacity constriction.* The capacity level shall be denoted by a constriction calibrated "to deliver" at 20°C when blown.

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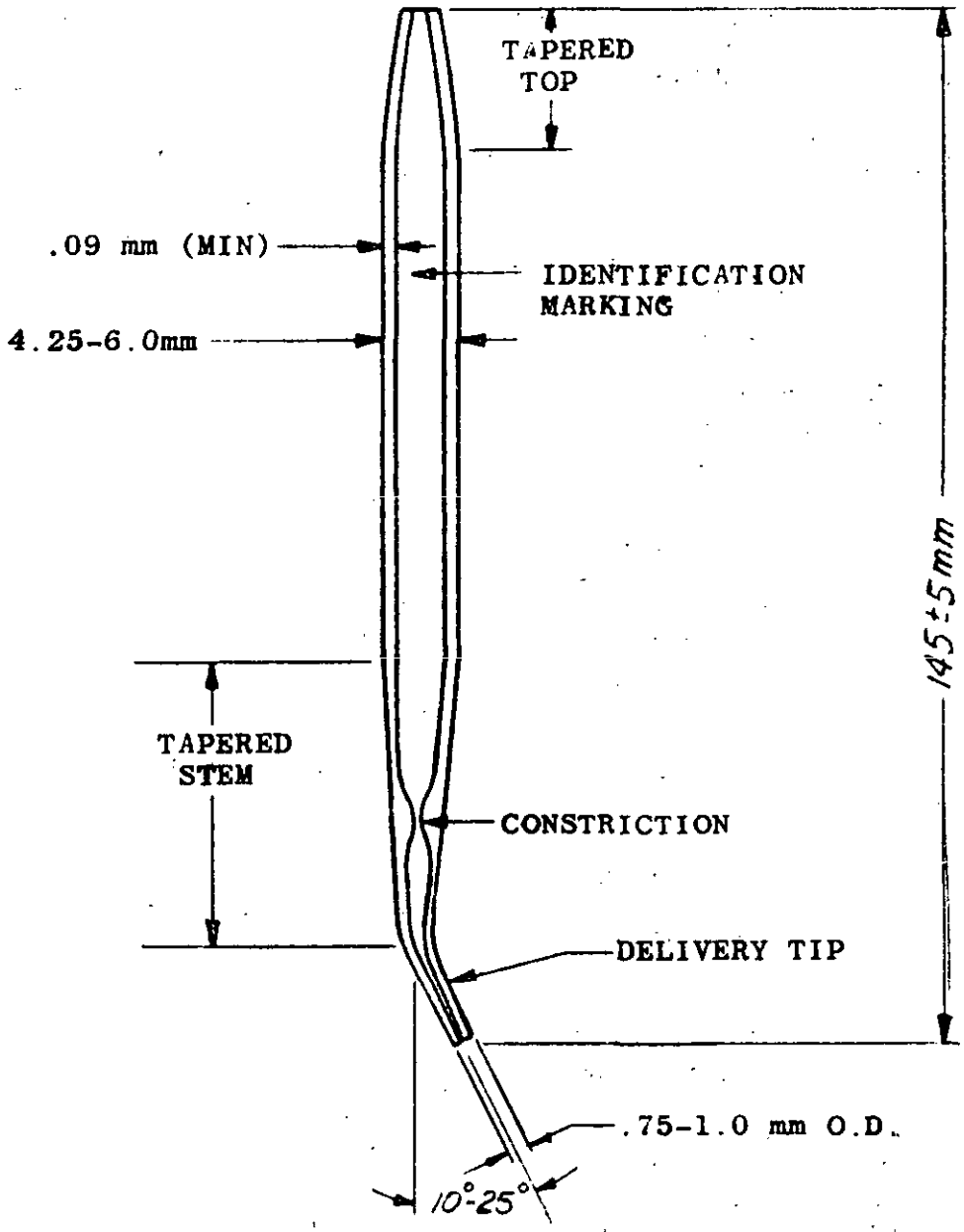


Figure 1—Lang-Levy pipets

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The outside diameter, at the constriction, may be indented.

3.4.2 *Delivery tip.* The delivery tip shall be tapered and bent at an angle of not less than 10 degrees or more than 25 degrees to a longitudinal axis of the micropipet. The end of the delivery tip shall be perpendicular to the longitudinal axis of the delivery tip and fire polished. Length of tip shall not exceed 10 mm measured from center of bend to end of tip.

3.4.3 *Bore diameters.* The bore diameter at the delivery tip, constriction, tapered top, and other points shall be such to deliver 2 milliliters of water in 60 to 120 seconds for the 1 to 10 microliter sizes and 25 to 60 seconds for the 15 to 500 microliter sizes when tested as specified in 4.6.3.

3.5 Identification marking. Each micropipet shall be permanently and legibly marked with the manufacturer's name or trademark and its capacity. Markings shall not change in appearance, as judged by the eye, when tested as specified in 4.6.4.

3.6 Workmanship. All micropipets shall be transparent and free from chips, cracks, and other defects which impair their serviceability. Internal surfaces shall be such that air bubbles do not form during filling.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 *Alternative inspection.* The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Objective evidence.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.3 Lotting. A lot shall consist of the micropipets of the same type and same capacity offered for acceptance at one time, which have been produced by one manufacturer under essentially the same manufacturing conditions.

4.4 Sampling.

4.4.1 *For examination.* Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2 *For test.* A sample of 3 micropipets shall be taken at random from each 100 micropipets or fraction thereof in the lot. Each micropipet in the sample shall be tested as specified in 4.6.1, 4.6.2, and 4.6.3 prior to testing as specified in 4.6.4. A different 1/3 of the sample shall be tested as specified in 4.6.4.1 and 4.6.4.2.1 and 4.6.4.2.2, respectively.

4.5 Inspection provisions.

4.5.1 *For examination.* Sample micropipets and the level A and level B preparation for delivery thereof shall be examined in accordance with the classification of defects and with MIL-STD-105.

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4.5.2 *For test.* Sample micropipets shall be tested in accordance with 4.6. The acceptance number is 0.

4.5.3 *Classification of defects.*

4.5.3.1 *Micropipets.*

Categories	Defects
Critical:	None defined.
Major:	*AQL 2.5 percent defective*
101	Shape not as specified.
102	Tip not as specified.
103	Dimension or constriction not as specified.
104	Damage (chips, cracks, etc.)
105	Identification marking illegible, incorrect, or incomplete.

4.5.3.2 *Preparation for delivery (section 5).*

Categories	Defects
Critical:	None defined.
Major:	*AQL 2.5 percent defective*
101	Unit, intermediate, or shipping container not as specified.
102	Number of micropipets per intermediate or shipping container not as specified or indicated.
103	Unlike micropipets packaged per intermediate or shipping container.
104	Closure of unit, intermediate, or shipping container not as specified.
105	Unit, intermediate, or shipping container damaged.
106	Marking of unit, intermediate, or shipping container illegible, incorrect, or incomplete.

4.6 **Tests.** Distilled water and reagent grade chemicals shall be used throughout the tests.

4.6.1 *Annealing.* Micropipets are tested in accordance with method A or B of ASTM Designation: C148.

4.6.2 *Capacity and tolerance.* After cleaning and drying, the micropipet and water for calibration are allowed to reach thermal equilibrium and the temperature of the water recorded. The micropipet is filled with water to the constriction (the excess water removed from the outside surfaces of the micropipet) and weighed on a microanalytical balance. The micropipet is drained, blown out, and reweighed. (The blowout is accomplished by a single strong puff.) The difference in weight is the weight of the water delivered. The capacity of the micropipet is calculated from the weight of water delivered using the "Relative Density and Volume of Water" table found in various laboratory handbooks. The capacity tolerance is calculated as follows:

$$\text{Capacity tolerance} = \frac{(V_c - V_L)\ 100}{V_L}$$

where: V_c = Calculated capacity of the pipet and

V_L = Listed capacity of the pipet.

(Note: See 6.5.)

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4.6.3 *Delivery time.* After cleaning, connect the top end of the micropipet to a 48-inch length of pure gum rubber tubing having a bore diameter of approximately 4 mm. Mount the micropipet in a vertical position so that the delivery tip is above the top end of the micropipet. Attach the other end of the rubber tubing to the delivery tip of a vertically mounted, clean buret conforming to NNN–B–789, style I, size 3. Fill the buret with water and open buret stopcock until the rubber tubing and micropipet are filled with water. Adjust the mountings so that there is a flow head of 36 ± ¼ inches between the water level in the buret and the delivery tip of the micropipet. Open wide the buret stopcock and measure, by means of a stopwatch, the time required for 2 ml of water to pass through the pipet.

4.6.4 *Permanency of marking.* A different micropipet shall be used for each of the following tests:

4.6.4.1 *Resistance to alkali.* The micropipet is completely immersed in 1*N* sodium hydroxide at room temperature and allowed to stand in the alkali for 1½ hours. The micropipet is then removed from the alkali, thoroughly rinsed with water, and dried by rubbing vigorously with a cloth.

4.6.4.2 *Resistance to acid.*

4.6.4.2.1 *Hydrochloric acid.* The micropipet is completely immersed in concentrated hydrochloric acid (specific gravity 1.18) at room temperature and allowed to stand in the acid for 1½ hours. The micropipet is then removed from the acid, thoroughly rinsed with water, and dried by rubbing vigorously with a cloth.

4.6.4.2.2 *Sulfuric acid.* The micropipet is completely immersed in concentrated sulfuric acid (specific gravity 1.84) at room temperature and allowed to stand in the acid for 1 hour. The micropipet is then removed from the acid, thoroughly rinsed with water, and dried by rubbing vigorously with a cloth.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Unit packaging.* Each micropipet shall be packaged in a clear plastic tube. Both ends of the tube shall be plugged with a soft resilient cushioning material.

5.1.1.2 *Intermediate packaging.* The number of unit packages specified (see 6.2), of the same capacity, shall be intermediately packaged in a snug-fitting box conforming to PPP–B–566, style I or II, type and class optional; PPP–B–636, type CF, class domestic; or PPP–B–676, type, class, and style optional. Box closure shall be secured with tape conforming to UU–T–111.

5.1.2 *Level C.* Micropipets shall be packaged in accordance with the manufacturer's commercial practice for shipment from supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Micropipets shall be packed in a box conforming to PPP–B–621, class 2 or PPP–B–601, overseas type. As far as practical, exterior containers shall be of uniform size and shape, be of minimum cube and tare, and contain identical quantities of like items. The gross weight of each container shall not exceed 150 pounds. Containers shall be closed and strapped in accordance with the applicable box specification and appendix thereto.

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5.2.2 *Level B.* Micropipets shall be packed in fiberboard boxes conforming to PPP–B–636, type CF or SF, class domestic, minimum bursting strength 275 pounds per square inch.

5.2.3 *Level C.* Micropipets shall be packed to insure carrier acceptance and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use. Micropipets covered by this specification are intended for laboratory use—to transfer single volumetric quantities in the range of 0.001 to 0.500 milliliter.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type and capacity required.

(*c*) Level of packaging and packing required.

(*d*) Number of unit packages per intermediate package required.

6.3 Military use. Items procured under this specification for military use are to be limited to the variety shown on the applicable military standard and military supply standard. Personnel of the military departments are requested to refer to these documents for guidance.

6.4 Illustrations. Figure 1 is for convenience of identification and to illustrate the characteristic features. It is not meant to limit procurement to that illustrated.

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6.5 Example of capacity tolerance.

At 23.3°C, 0.00381 gram of water discharged from a 4-microliter pipet.
—From handbook table, 1 ml of water weighs 0.99749 gram at 23.3°C.

$$V_c \text{ (Calculated capacity)} = \frac{0.00381}{0.99749} = 0.0038195 = 3.82 \text{ microliters.}$$

$$\text{Capacity tolerance} = \frac{(V_c - V_L)\ 100}{V_L} = \frac{(3.82 - 4.00)\ 100}{4.00} = -4.5 \text{ percent.}$$

The capacity tolerance is in slight error in that the calculated capacity (V_c) is at 23.3°C rather than the specified 20°C. However, the correction is negligible for the purpose of the calculation at this particular temperature. To convert V_c @ 23.3°C to V_c @ 20°C, the following equation is applicable (reference: NBS Circular 602):

$$V_{20} = \frac{V_t}{1 + \alpha\ (t\text{-}20)}$$

where,
$V_{20} = V_c$ @ 20°C,
$V_t = V_c$ @ 23.3°C = 3.8195 (or 3.82) microliters,
$t = 23.3$°C, and
$\alpha = 0.000010$ for borosilicate glass and
0.000025 for soda lime glass.

then,
for type I (borosilicate glass) micropipets,

$$V_c \text{ @ } 20°C = \frac{3.8195}{1 + 0.00001\ (23.3 - 20.0)} = 3.8193 = 3.82 \text{ microliters}$$

and for type II (soda-lime glass) micropipets,

$$V_c \text{ @ } 20°C = \frac{3.8195}{1 + 0.000025\ (23.3 - 20.0)} = 3.8191 = 3.82 \text{ microliters.}$$

6.6 Transportation description. Transportation description applicable to this item is:

Glassware, chemical laboratory.

Carload minimum weight 18,000 pounds, subject to Rule 34, Uniform Freight Classification.

Glassware, chemical laboratory.

Truckload minimum weight 18,000 pounds, subject to Rule 115, National Motor Freight Classification.

CUSTODIANS:

Army—MU

Navy—BuMed

Air Force—MAA

Preparing activity:

Army—MU

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